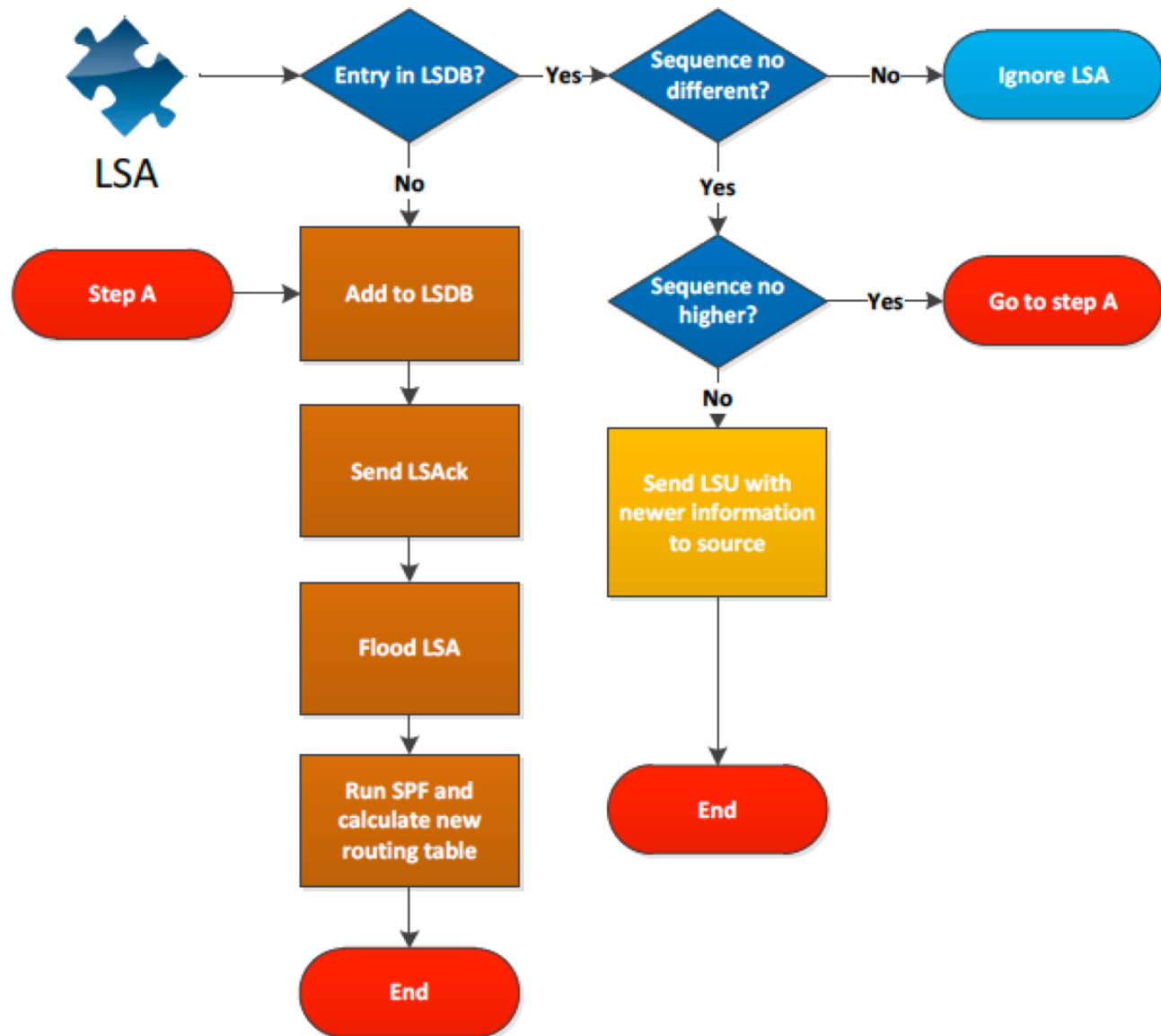


LSA Updates

- link-state routing protocols generate routing updates **only** when a **change** occurs in the network topology.
- when a **link changes state**, the device that detected the change creates a **link-state advertisement** (LSA) concerning that link and sends it to **all neighboring devices** using a special multicast address.
- each routing device reads the LSA.
 - the LSA has a **sequence number** that allows the router to check to see if it has already seen that update
 - if **old**, it is **discarded**, if **new**, link-state database (LSDB) info **updated** and LSA passed along to next neighbors

Flow Chart

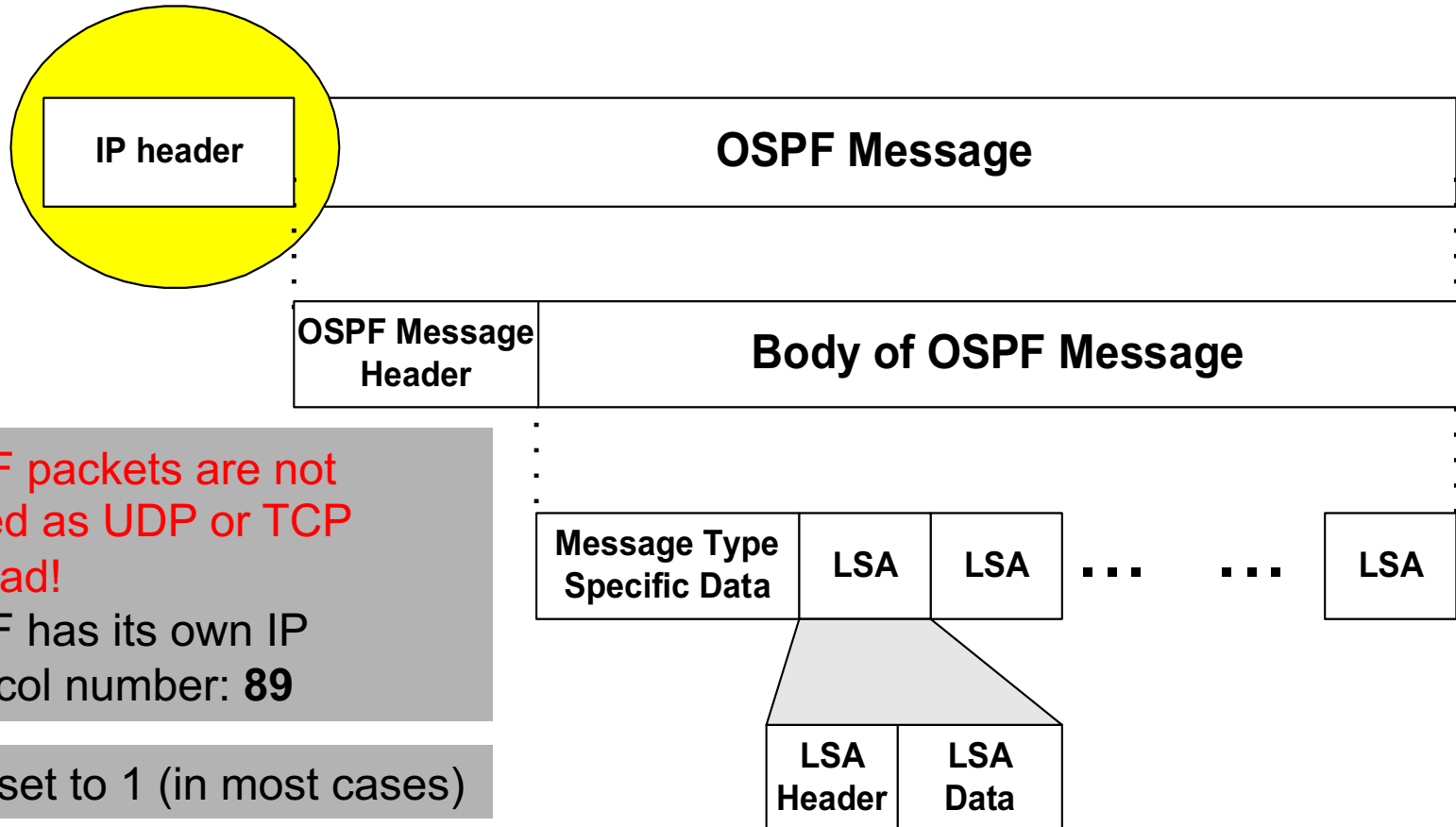


OSPF Link State Packets

There are **five** types of Link-State Packets (LSPs).

1. **hello**: are used to establish and maintain adjacency with other OSPF routers. They are also used to elect the Designated Router (DR) and Backup Designated Router (BDR) on multi-access networks.
2. **database description (DBD or DD)**: contains an **abbreviated list** of the sending router's link-state database and is used by receiving routers to check against the local link-state database to make sure it has the latest information.
3. **link-state request (LSR)**: used by routers to **request** more information about any entry in the DBD
4. **link-state update (LSU)**: used to **reply** to LSRs as well as to announce new information. LSUs can contain 7 different types of Link-State Advertisements (**LSAs**)
5. **link-state acknowledgement (LSAck)**: sent to **confirm** receipt of an LSU message

OSPF Packet Format



OSPF packets are not carried as UDP or TCP payload!

OSPF has its own IP protocol number: **89**

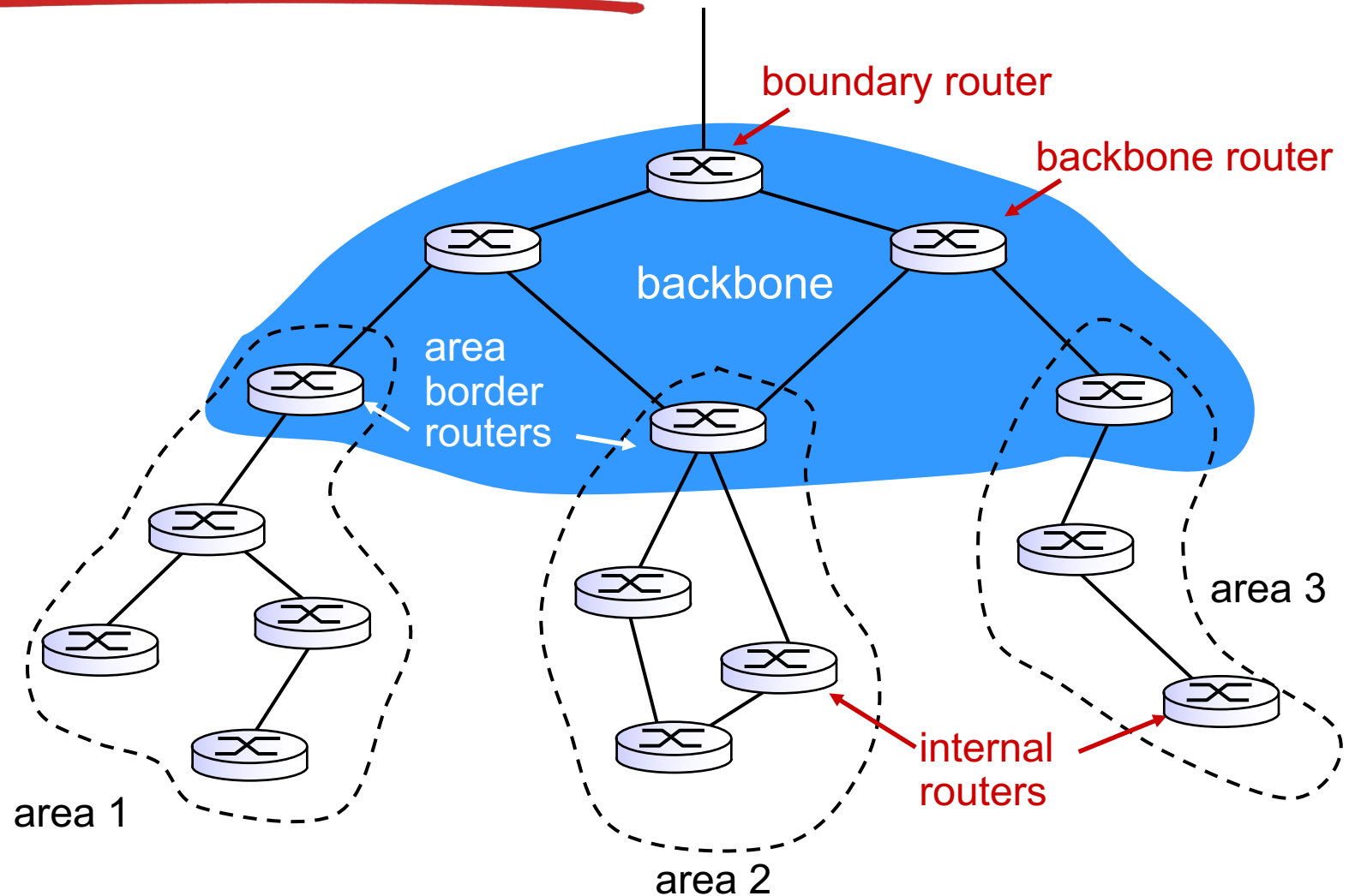
TTL: set to 1 (in most cases)

Destination IP: neighbor's IP address or 224.0.0.5 (ALLSPFRouters) or 224.0.0.6 (AllDRouters: (designated and backup designated only)

OSPF “advanced” features

- **security**: all OSPF messages authenticated (to prevent malicious intrusion)
- **multiple** same-cost **paths** allowed (only one path in RIP)
- for each link, multiple cost metrics for different **TOS** (e.g., satellite link cost set low for best effort ToS; high for real-time ToS)
- integrated uni- and **multi-cast** support:
 - Multicast OSPF (MOSPF) uses same topology data base as OSPF
- **hierarchical** OSPF in large domains.

Hierarchical OSPF



Hierarchical OSPF

- *two-level hierarchy*: local area, backbone.
 - link-state advertisements only in area
 - each nodes has detailed area topology; only know direction (shortest path) to nets in other areas.
- *area border routers*: “summarize” distances to nets in own area, advertise to other Area Border routers.
- *backbone routers*: run OSPF routing limited to backbone.
- *boundary routers*: connect to other AS' es.

Chapter 5: outline

5.1 introduction

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet:

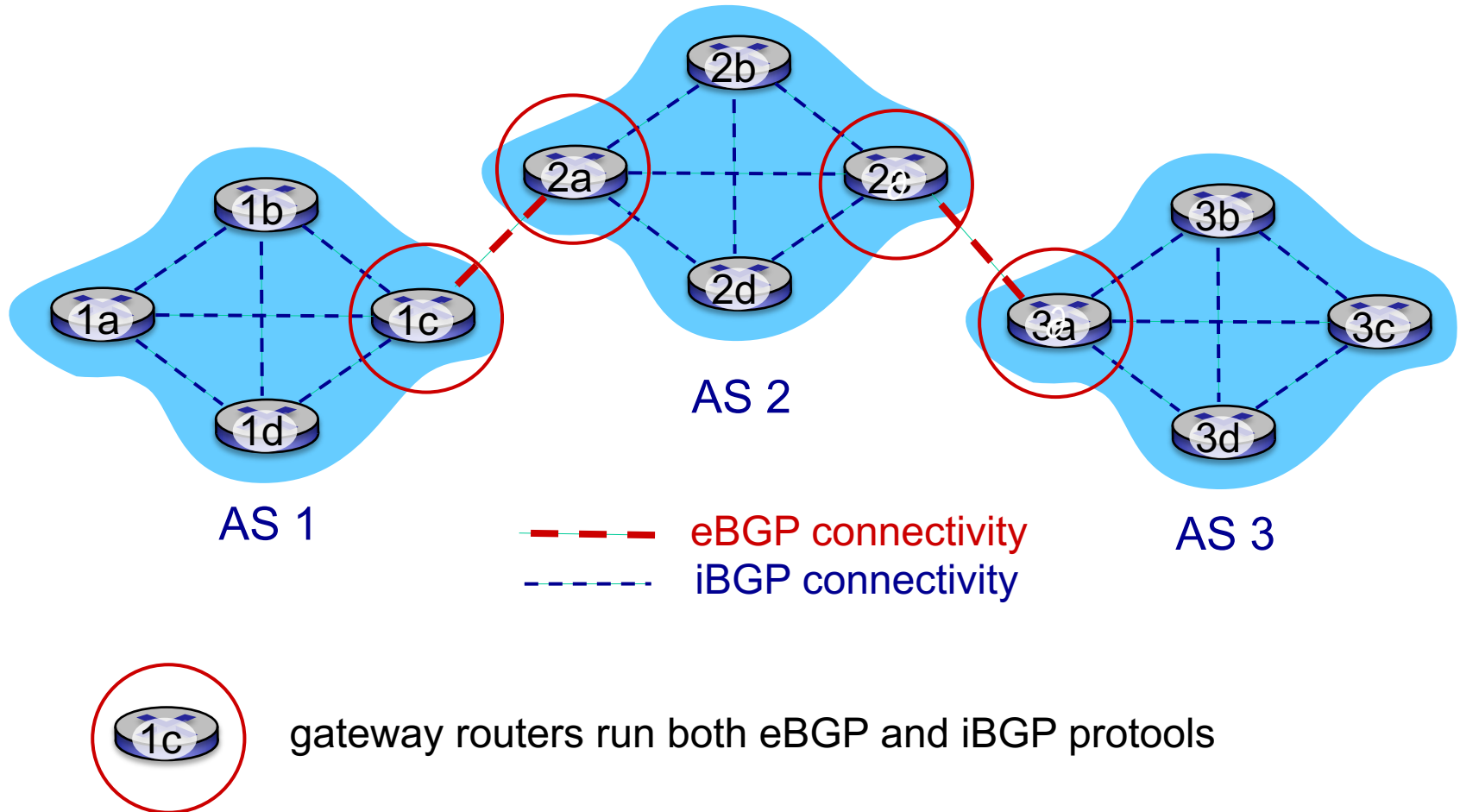
- RIP
- OSPF

5.4 inter-AS routing in the Internet BGP

Internet inter-AS routing: BGP

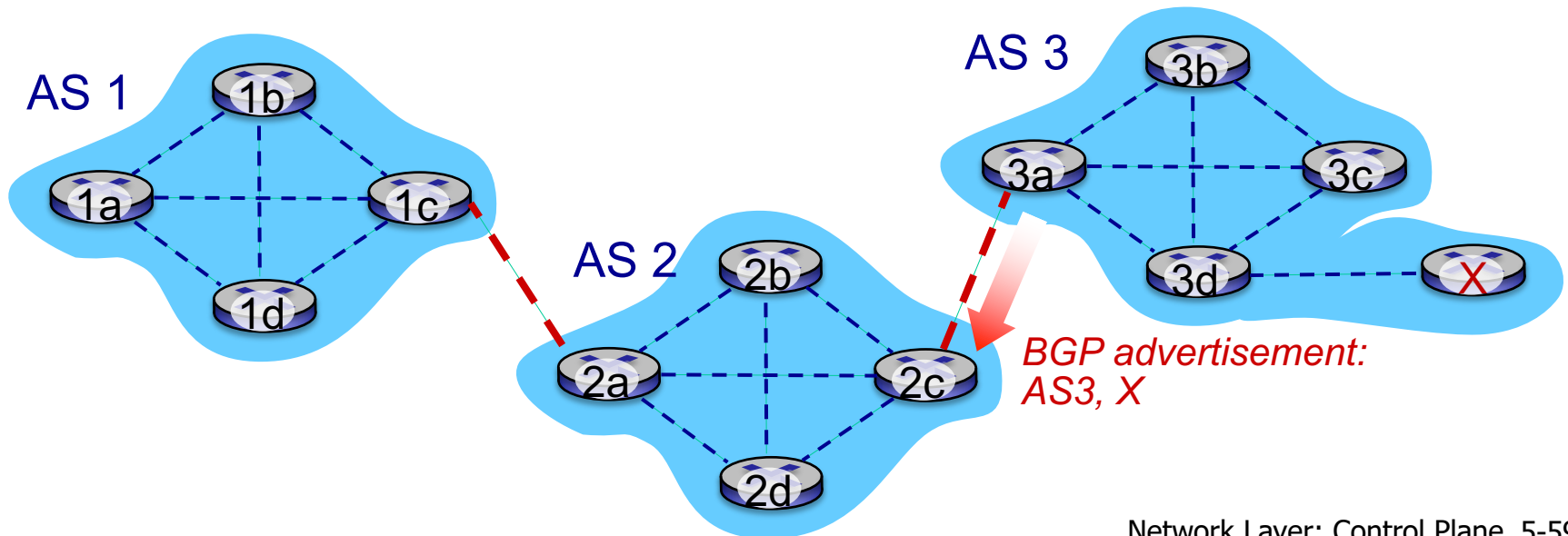
- **BGP (Border Gateway Protocol):** *the de facto inter-domain routing protocol (v 4)*
 - “glue that holds the Internet together”
- BGP provides each AS a means to:
 - **eBGP:** obtain subnet reachability information from neighboring ASes
 - **iBGP:** propagate reachability information to all AS-internal routers.
 - determine “good” routes to other networks based on **reachability** information and **policy**
- allows subnet to advertise its existence to rest of Internet: “*1 am here*”
- uses TCP for **reliable** communications to transmit routing messages

eBGP, iBGP connections



BGP basics

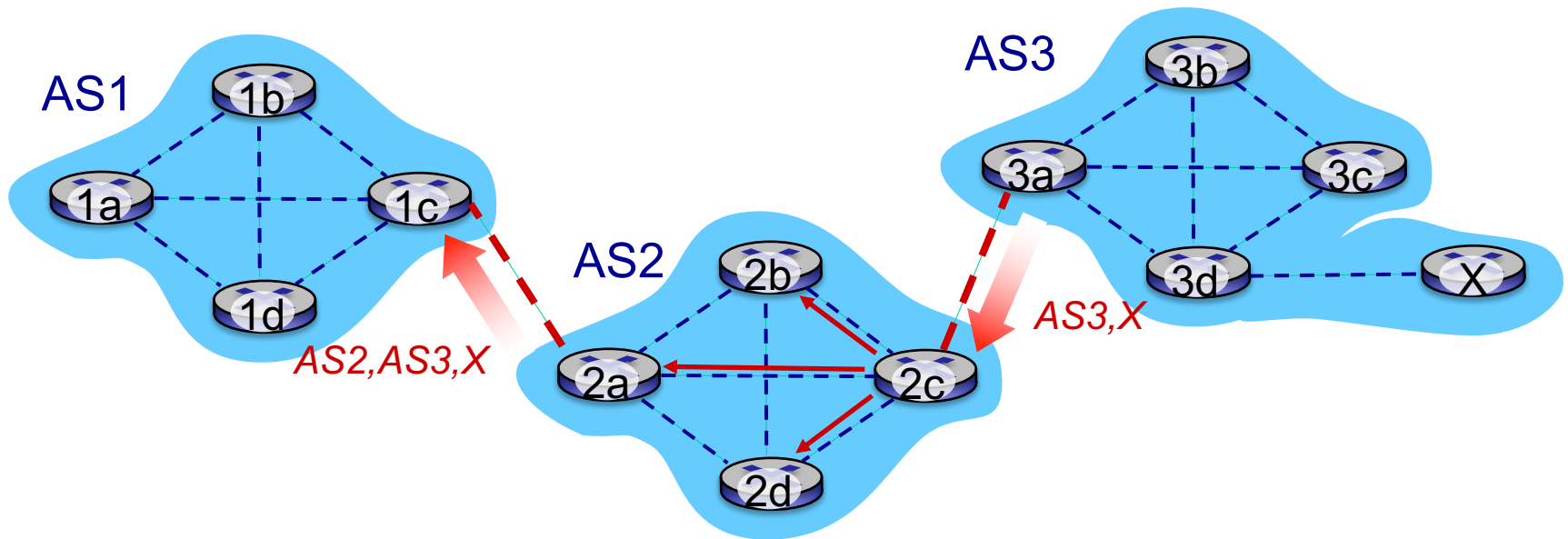
- **BGP session:** two BGP routers (“peers”) exchange BGP messages over semi-permanent TCP connection:
 - advertising *paths* to different destination **network prefixes** (BGP is a “path vector” protocol)
- when AS3 gateway router 3a advertises path **AS3,X** to AS2 gateway router 2c:
 - AS3 *promises* to AS2 it will forward datagrams towards **X**



Path attributes and BGP routes

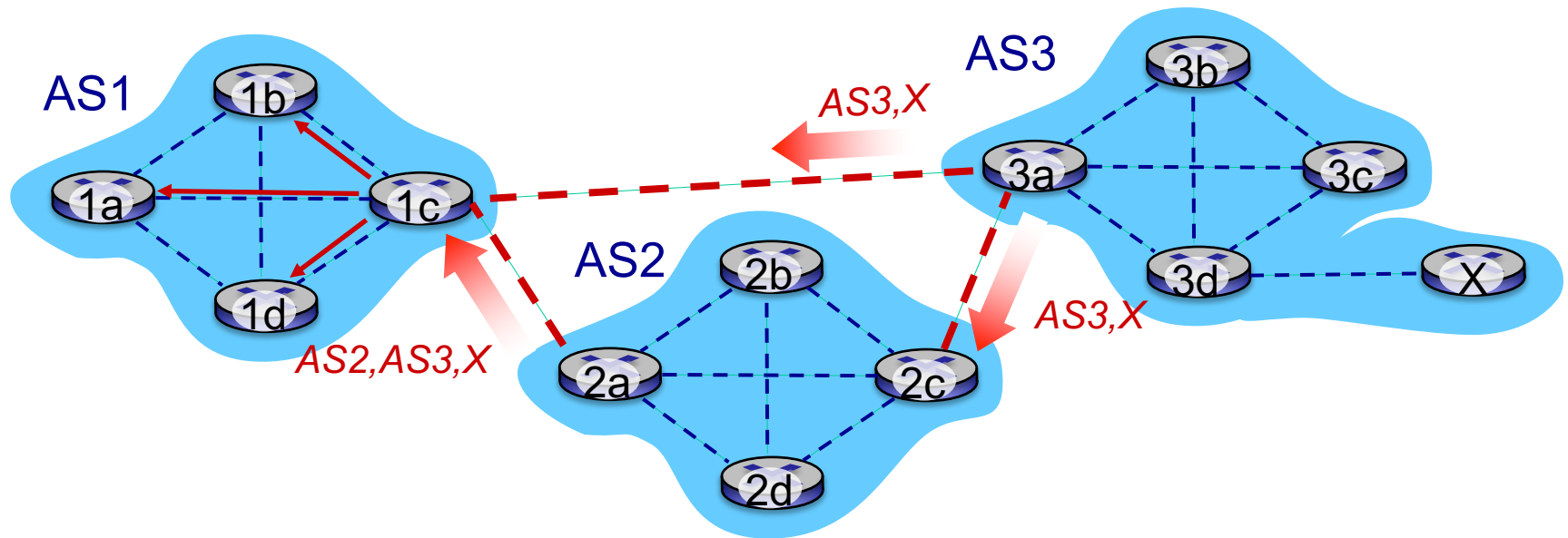
- advertised prefix includes BGP attributes
 - prefix + attributes = “route”
- two important attributes:
 - **AS-PATH**: list of ASes through which prefix advertisement has passed
 - **NEXT-HOP**: indicates specific internal-AS router to next-hop AS
- *Policy-based routing*:
 - gateway receiving route advertisement uses *import policy* to accept/decline path (e.g., never route through AS Y).
 - AS *export policy* also determines whether to *advertise* path to other neighboring ASes

BGP path advertisement



- AS2 router 2c receives path advertisement **AS3,X** (via eBGP) from AS3 router 3a
- Based on AS2 **import policy**, AS2 router 2c **accepts** path AS3,X, and propagates (via iBGP) to all AS2 routers
- Based on AS2 **export policy**, AS2 router 2a **advertises** (via eBGP) path **AS2,AS3,X** to AS1 router 1c

BGP path advertisement

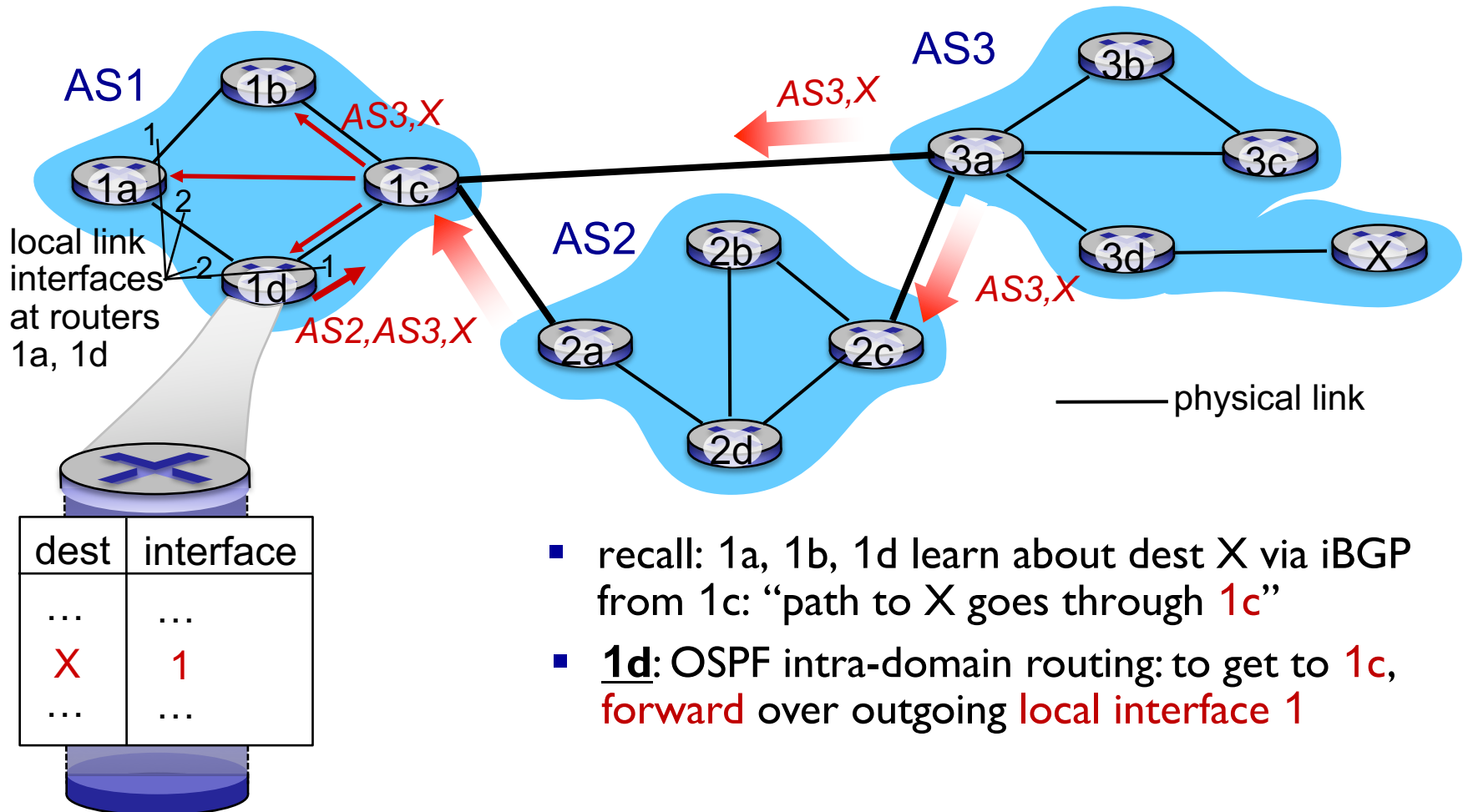


gateway router may learn about **multiple** paths to destination:

- AS1 gateway router 1c learns path *AS2,AS3,X* from 2a
- AS1 gateway router 1c learns path *AS3,X* from 3a
- Based on policy, AS1 gateway router 1c chooses path *AS3,X*, and *advertises path within AS1 via iBGP*

BGP, OSPF, forwarding table entries

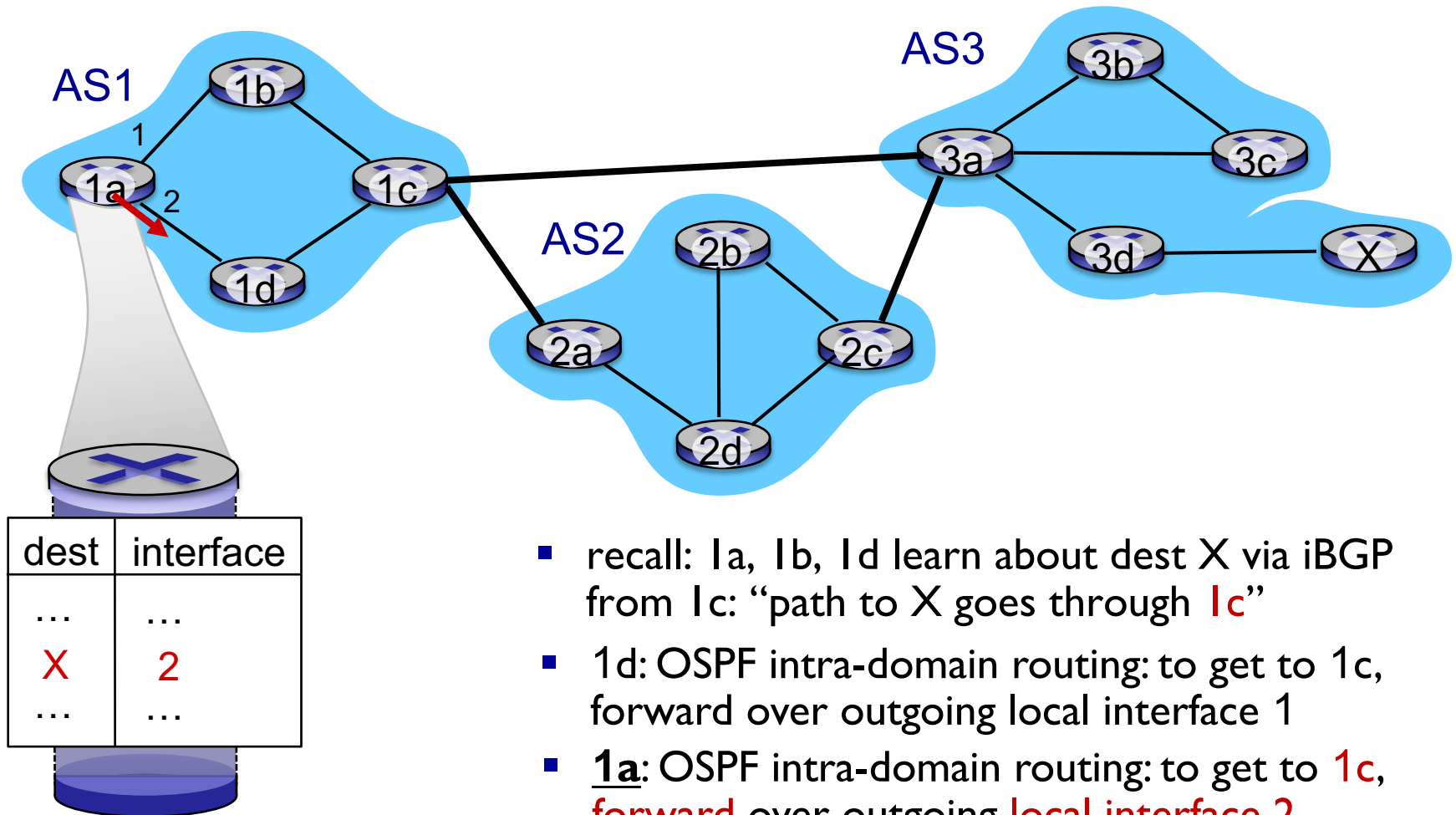
Q: how does router set forwarding table entry to distant prefix?



- recall: 1a, 1b, 1d learn about dest X via iBGP from 1c: “path to X goes through **1c**”
- 1d**: OSPF intra-domain routing: to get to **1c**, **forward** over outgoing **local interface 1**

BGP, OSPF, forwarding table entries

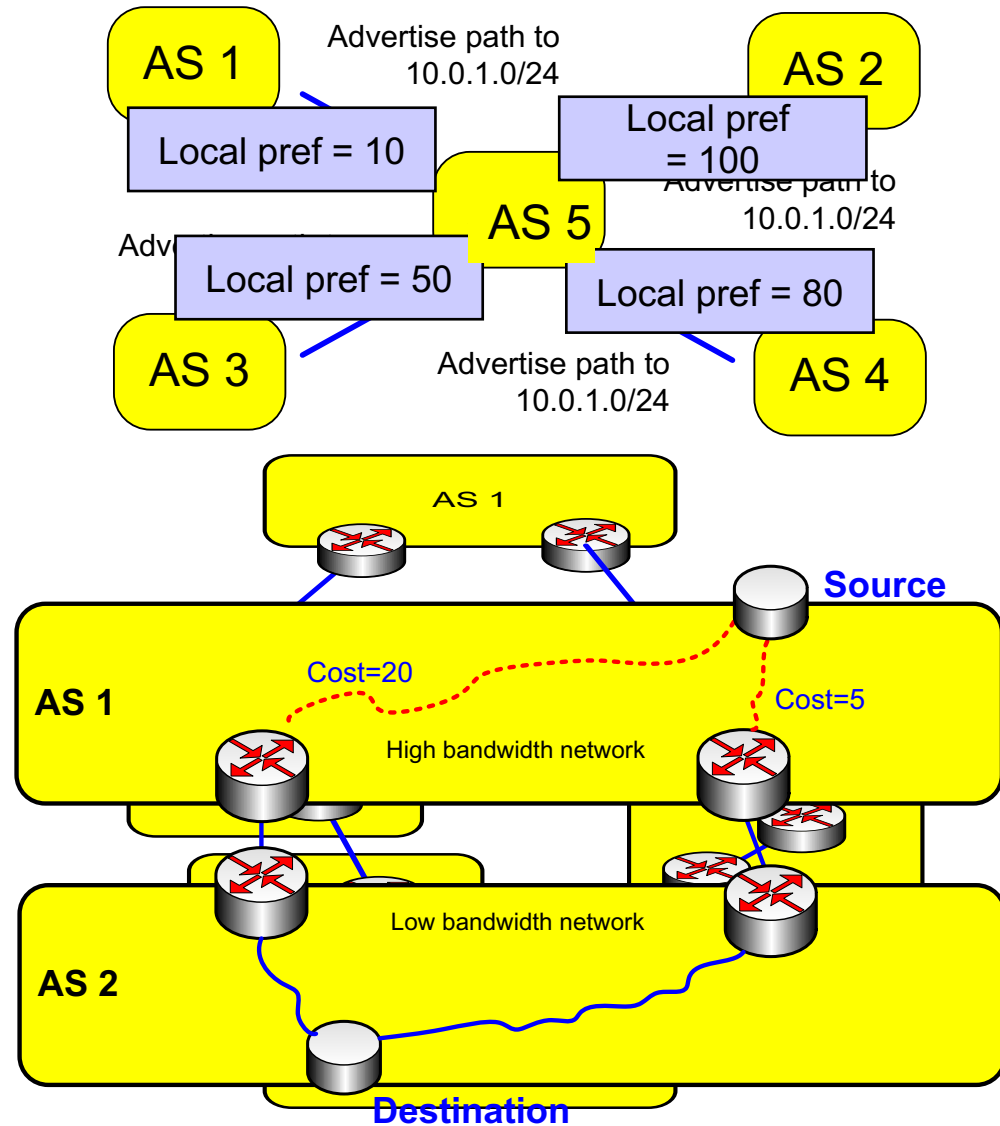
Q: how does router set forwarding table entry to distant prefix?



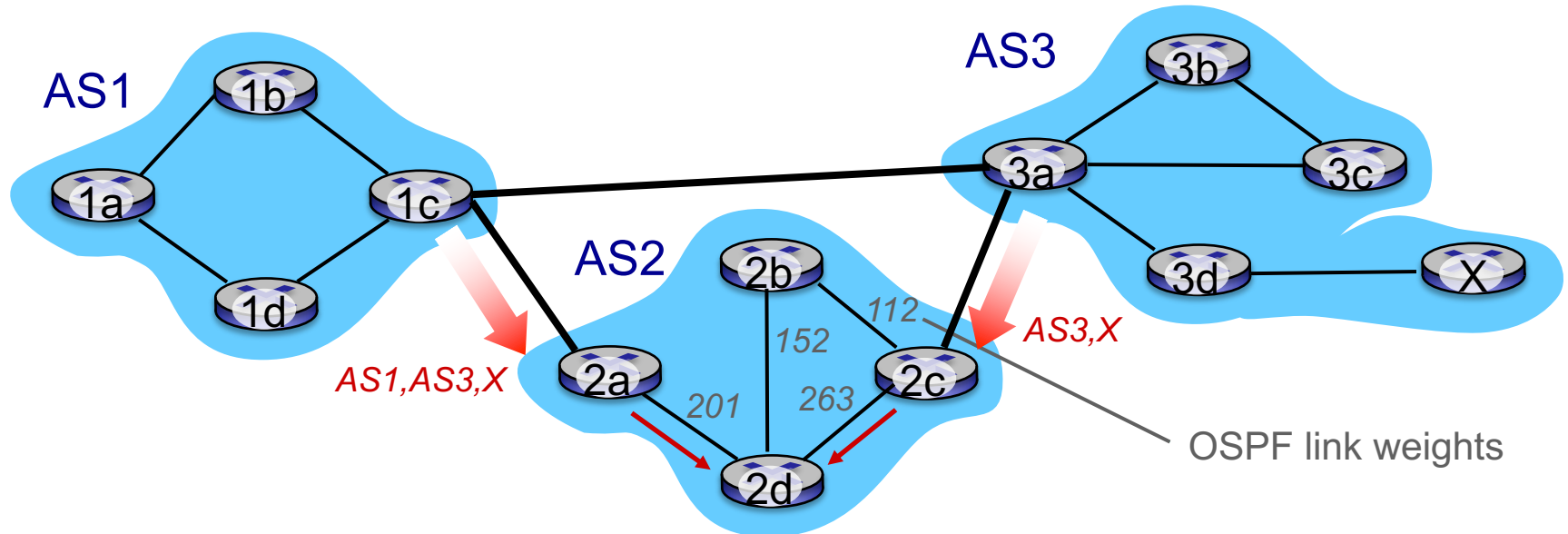
- recall: 1a, 1b, 1d learn about dest X via iBGP from 1c: “path to X goes through **1c**”
- 1d: OSPF intra-domain routing: to get to 1c, forward over outgoing local interface 1
- 1a: OSPF intra-domain routing: to get to **1c**, forward over outgoing **local interface 2**

BGP route selection

- gateway router may learn about more than one route to destination AS, selects route based on:
 - local preference value attribute: **policy decision**
 - shortest AS-PATH
 - closest NEXT-HOP internal router: hot potato routing
 - additional criteria
- shortest AS-PATH may not mean shortest router/hop path
- best cost intra path may not mean best cost overall



Hot Potato Routing



- 2d learns (via iBGP) it can route to X via 2a or 2c
- *hot potato routing*: choose local gateway that has least intra-domain cost (e.g., 2d chooses 2a, even though more AS hops to X): don't worry about inter-domain cost!
- Note that BGP will go with shortest AS path first. If 2 routes have **same length AS path**, use hot potato (shortest path) internally (IGP).